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Magmatism of the mid-Proterozoic Gardar Province, South Greenland: chronology, petrogenesis and geological setting

B.G.J. Upton[a,*], C.H. Emeleus[b], L.M. Heaman[c], K.M. Goodenough[d], A.A. Finch[e]

[a] *Department of Geology and Geophysics, University of Edinburgh, King's Building, West Mains Road, Edinburgh EH9 3JW, UK*
[b] *Department of Geological Sciences, University of Durham, Durham DH1 3LE, UK*
[c] *Department of Earth and Atmospheric Sciences, University of Alberta, Edmonton, Alberta, Canada T6G 2E3*
[d] *British Geological Survey, Murchison House, Edinburgh EH9 3LA, UK*
[e] *School of Geography and Geosciences, University of St. Andrews, St. Andrews KY16 9AL, UK*

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Abstract

A lithospherically weak zone embracing the southern margin of the Greenland Archaean craton and the Ketilidian Julianehåb batholith underwent repeated rifting during the interval 1350–1140 Ma, accompanying breakup of Palaeopangaea. The Gardar Province comprises the area affected by the faulting and asssociated alkaline magmatism. While an estimated 2–5 km of Proterozoic cover has been eroded, rift-fill successions have been preserved in early fault-bounded basins.

The orientation of dyke swarms changed from WNW–ESE to nearly NE–SW during Gardar times. The principal swarms are inferred to occupy zones of lithospheric thinning and graben development. Central-type intrusive complexes, largely of syenites and nepheline syenites, reached shallow levels. The principal magmatic evolution was from transitional olivine basalt through to phonolites. Accompanying silica-oversaturated magma generation involved greater degrees of crustal assimilation. Anorthositic xenoliths in the Gardar intrusions imply the presence of an extensive anorthositic complex at depth, regarded as an integral part of the North American Proterozoic massif anorthosite association. The most primitive Gardar basalts are themselves relatively evolved, probably as a result of olivine ± pyroxene fractionation during crustal underplating. The Gardar basic rocks are troctolitic with elevated Al_2O_3/CaO ratios: their incompatible element patterns suggest a significant input from lithospheric sources. The Ca-deficient nature of the Gardar basalts is attributed to an origin involving lithospheric mantle depleted by previous melting events. Trace element and isotopic signatures suggest considerable heterogeneity in the mantle sources which are ascribed to differential metasomatism of clinopyroxene-poor peridotite sources by small-fraction asthenospheric melts. The ultramafic lamprophyre/carbonatite association that recurred throughout the period is inferred to have originated from melting of metasomites deep within the lithospheric mantle. Affinities between the alkaline intrusions over an interval of >100 Ma characterize the Gardar as a coherent magmatic province and support the contention that the magmas are largely of lithospheric origin. The energy required to generate the very large requisite volumes of primitive magmas may have been supplied by successive mantle plumes. The Gardar magmatism pre- and post-dates the ~1.27 Ga Mackenzie Igneous Events of North America but wholly pre-dates the ~1.1 Ga Keweenawan magmatism associated with the Midcontinent Rift.
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Keywords: Greenland; Continental rifting; Mesoproterozoic; Alkaline magmatism

* Corresponding author. School of Geosciences, Grant Institute, University of Edinburg, West Mains Road, Edinburgh EH9 3JW, UK. Tel.: +44-31-650-7285; fax: +44-31-668-3184.
E-mail address: bupton@glg.ed.ac.uk (B.G.J. Upton).

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1. Introduction

Wegmann (1938) adopted the term 'Gardar Formation' from the old Norse bishopric of Gardar in south Greenland, as the name for the distinctive association of essentially unmetamorphosed and undeformed alkaline intrusive rocks and successions of clastic sediments and subaerial lavas which are beautifully exposed in the area. Earlier reviews of the province have been presented by Upton (1974), Emeleus and Upton (1976), Upton and Emeleus (1987) and Kalsbeek et al. (1990), and the justification for a further review stems from the rapid acquisition of new data over recent years and their consideration in terms of new concepts. The paper examines the geodynamic aspects of the Gardar magmatism within the broader context of its relationship to the large igneous provinces developed within North America during the mid-Proterozoic.

The intrusions principally comprise dykes (mainly mildly alkalic or 'transitional' dolerites) and subcylindrical plutons (mainly syenites). Detailed mapping has shown that the magmatism was accompanied by faulting (Emeleus, 1964; Berthelsen and Noe-Nygaard, 1965; Watt, 1968; Stephenson, 1976). Rb–Sr age-dating, mostly initiated by A.B. Blaxland, demonstrated that the Gardar intrusions are Mesoproterozoic with a suggested age range from ~1350 to 1160 Ma (Blaxland et al., 1978).

Three principal episodes of rift-related magmatism were distinguished by Upton and Blundell (1978), occurring at approximately 1350–1300, 1280–1260 and 1180–1140 Ma. This work has since been corroborated and amplified, particularly by more precise zircon and baddeleyite U–Pb dating. The new U–Pb age determinations have demonstrated at least two periods of igneous activity at ~1280 Ma and

Table 1
Compilation of U–Pb radiometric dates from Gardar igneous rocks

Unit	Location	Age (Ma)	Reference
BD_0 dyke	Qaqortoq	1284 ± 3	1
BD_0 dyke	Kangerluarssuk	1280 ± 3	1
BD_0 dyke	Tugtutôq	1279 ± 1.3	1
Syenite pegmatite	Motzfeldt	1275.3 ± 1.1	2
Gabbro ring dyke	Kûngnât	1275.2 ± 1.8	1
Older giant dyke	Tugtutôq	1184 ± 5	1
Syenite pegmatite	Nunarssuit	1171 ± 5	3
Granite ('green') sheet	Ilímaussaq	1166 ± 9	1
Syenite pluton	Klokken	1166 ± 3	4
Younger giant dyke	Tugtutôq	1165.7 ± 1.2	1
Younger giant dyke	Tugtutôq	1163 ± 2	5
Agpaite cumulate	Ilímaussaq	1160 ± 5	6
Granite pluton	Tugtutôq	1156 ± 1.1	1
Syenite intrusion	Østfjordsdal	1147.5 ± 3.2	2
Syenite pluton	Paatusoq	1144 ± 1	5

Errors are quoted at the 2σ level.
References: (1) Heaman and Upton (in preparation); (2) Salmon et al. (in preparation); (3) Finch et al. (2001b); (4) Harper (1988); (5) M.A. Hamilton (personal communication); (6) Waight et al. (2002).

Table 2
Published Rb–Sr radiometric age dates from Gardar igneous rocks

Rock types	Location	Age (Ma)	Reference
Nepheline syenites and carbonatite	Grønnedal–Ika	1299 ± 17	1
Gabbro, nepheline syenite, lujavrite	Early Motzfeldt Complex	1282 ± 30	1
Nepheline syenites and lujavrite	North Qôroq	1268 ± 60	1
Lamprophyres and dolerites	Ivigtut region	1250 ± 18 1227 ± 29	2
Nepheline syenites	North Motzfeldt	1226 ± 27	3
Alkali granite and cryolite body	Ivigtut	1222 ± 24	1
Gabbros and syenites	Kûngnât	1219 ± 16	1
Trachytes and carbonatites	Qassiarsuk	1205 ± 12	4
Gabbros and syenites	Bangs Havn	1185 ± 22	5
Agpaite	Ilímaussaq	1160.1 ± 2.3	6
Syenites and nepheline syenites	South Qôroq	1160 ± 8	1
Gabbros, syenites and nepheline syenites	Tugtutôq (Older Giant Dyke)	1150 ± 9	1
Quartz syenites and granites	Tugtutôq (Central Complex)	1143 ± 36	1
Agpaites, syenites and granites	Ilímaussaq	1143 ± 21	1
Gabbros, syenites and nepheline syenites	Igdlerfigssalik ('Late Complex')	1142 ± 15	1
Gabbros and syenites	Klokken	1135 ± 11	1
Quartz syenites and granites[a]	Nunarssuit	1130 ± 14	1

Data are presented as simple regressions, recalculated for the decay constant of Steiger and Jäger (1977); errors are shown at the 2σ level.
References: (1) Blaxland et al. (1978); (2) Patchett (1977); (3) Finch et al. (2001b); (4) Andersen (1997); (5) Engell and Pedersen (1974); (6) Waight et al. (2002).
[a] Reference 1 provided regressions for four units within the Nunarssuit complex. They all lie within error of the value for the Nunarssuit syenite, which is quoted here.

between 1180 and 1140 Ma (Table 1). However, U–Pb zircon/baddeleyite dates have not yet been obtained for the early Gardar volcanic rocks of the Eriksfjord Formation nor for those plutons whose field-relations show them to have been the oldest. Consequently, the time at which Gardar activity began is still not well defined. The total age range for what may be termed the Gardar Period is thus at least 140 Ma but could be in excess of 200 Ma, approximately from 1350 to 1140 Ma. Published Rb–Sr ages are collated in Table 2.

The province developed in a continental environment that was subjected to transtensional tectonics and concomitant magmatism. These processes affected a terrain that had been deformed, metamorphosed and intruded during the Ketilidian orogeny (1900–1700 Ma) and which had been stabilised, uplifted and deeply eroded in the intervening 500–400 Ma. Ketilidian subduction of oceanic crust beneath the Archaean craton gave rise to orogenesis and emplacement of the I-type granitic Julianehåb batholith in a sinistral transpressional system (Garde et al., 2002; Chadwick et al., 1994). The Gardar Province mostly lies within the confines of this batholith, although its northern part transgresses the Border and Foreland Zones of the Archaean craton (Allaart, 1976). Most of the subsequent Gardar activity was localised within the confines of this batholith, especially along its axis. The Gardar structures were, to a large extent, controlled by the earlier Ketilidian structural patterns (Fig. 1).

The Gardar Province was, until recently, taken to be some 70 km wide. However, the recognition of the Paatusoq Complex, also lying within the Julianehåb

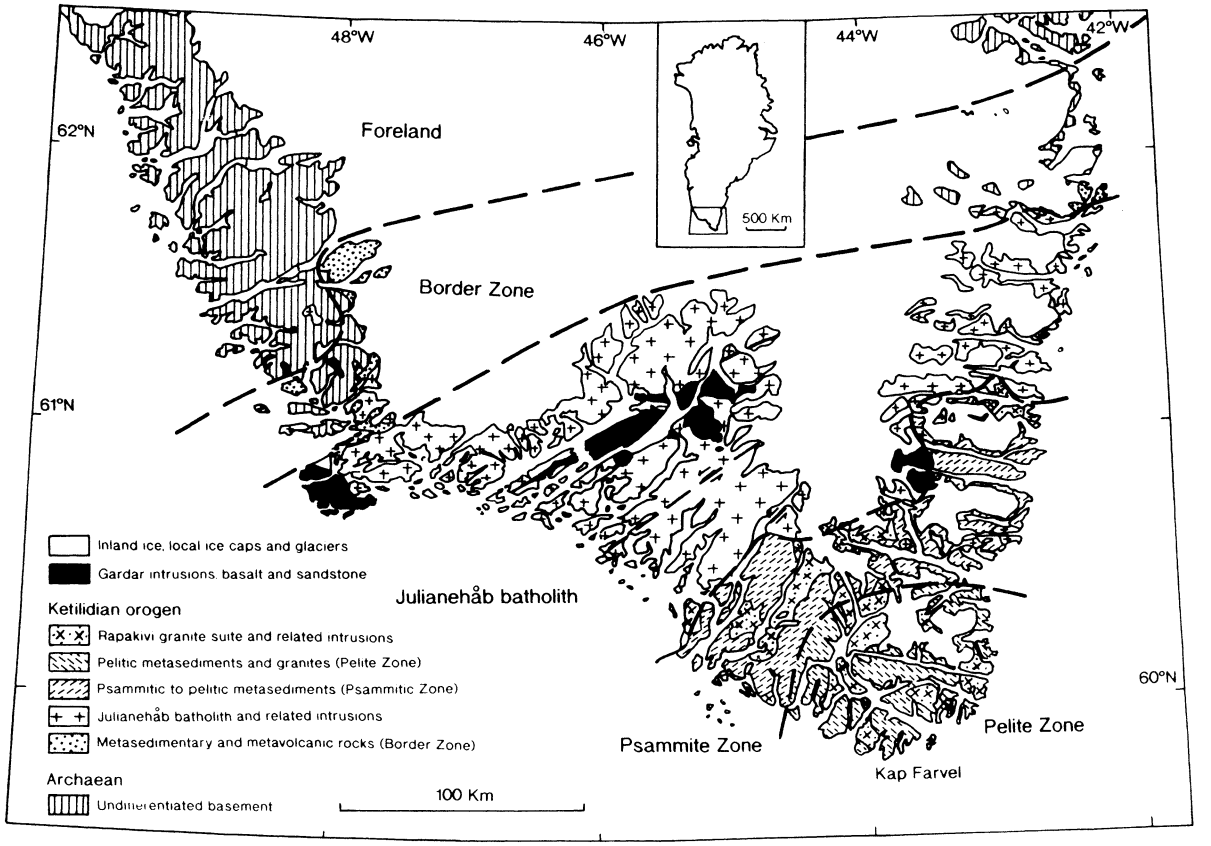


Fig. 1. Geological sketch map of southern Greenland showing the distribution of craton and zonation of the Ketilidian. The relationship of the Gardar rocks to the Border Zone and Ketilidian batholith is shown. The Paatusoq Complex is the Gardar intrusion on the eastern side of the map. (after Garde et al., 2002).

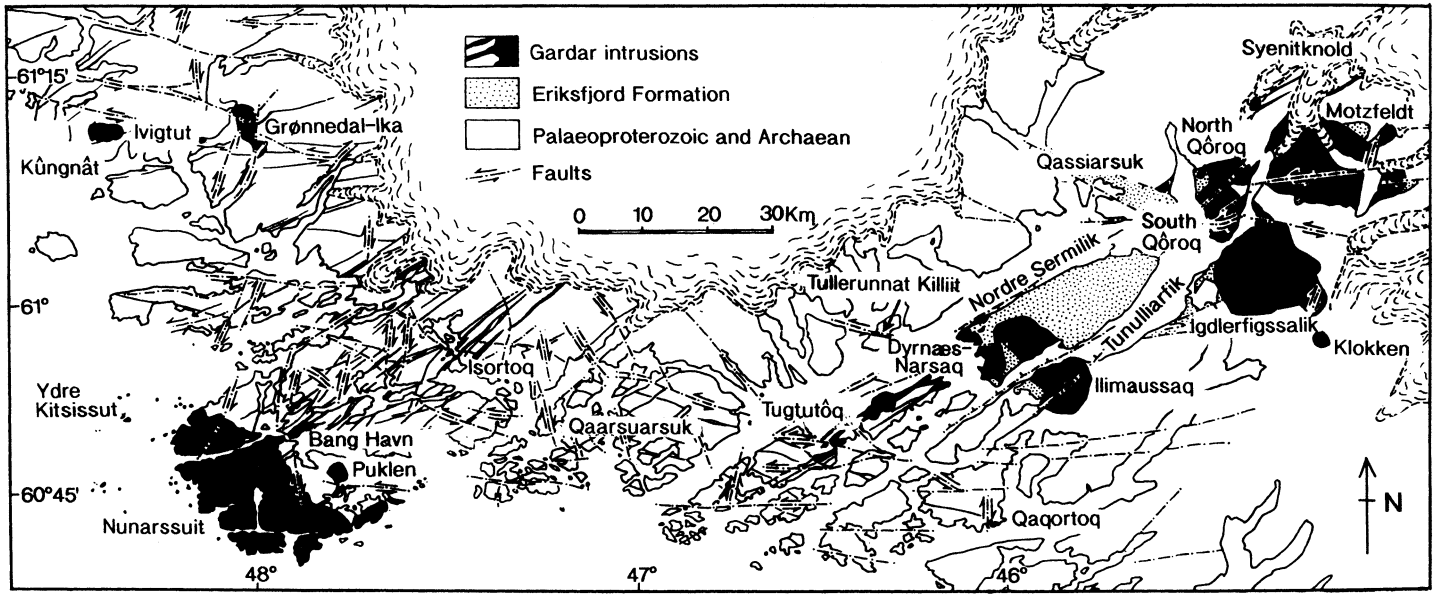


Fig. 2. Sketch map of the Gardar Province showing principal plutonic centres, dykes and the main outcrops of the Eriksfjord Formation. (after Upton, 1974). The nunataks referred to in the text lie north and northeast of Motzfeldt. The Motzfeldt, Qôroq and Igdlerfigssalik complexes are collectively termed the 'Igaliko Syenite Complexes'.

batholith but adjacent to its southern margin (Chadwick and Garde, 1996; Grocott et al., 1999; Garde et al., 2002), requires the province width to be increased to ~120 km (Fig. 1). It is concluded that an ENE-trending zone of inherent mechanical weakness adjacent to, and overlapping, the craton, largely coincident with the Julianehåb batholith, was repeatedly exploited by fracturing and magma ascent during the whole period of Gardar igneous activity. A pronounced elongate 'gravity high' (up to 300 g.u.) occurs parallel to, and almost coincident with, part of the zone of dyke and pluton intrusion lying along the axis of the Julianehåb batholith. This 'high' is believed to be due to a major basic intrusive measuring 80×25 km, rising close to surface level near Narsaq but lying at greater depths to the WSW (Blundell, 1978).

Most of the Gardar magmas ranged from transitional basalt to trachyte, with divergence towards either phonolitic or comenditic residua (Upton, 1974). Although the igneous rocks display a compositional continuum, there is a relative scarcity in terms of outcrop area of intermediate compositions (notably in the mugearite range) that may reflect a reduction in volumes of such magmas reaching high levels. The province thus exhibits a classic 'Daly gap', probably attributable to increased magma densities in the intermediate ranges (Watt, 1966; Upton, 1974).

In addition to the dominant (relatively silicic) mafic to salic rocks, very silica-deficient lamprophyres and carbonatites occur as minor components in the craton Border Zone and along the axis of the Julianehåb batholith (henceforward referred to as the batholith-axial zone). In the former area these are confined to intrusions while in the latter both the carbonatites and lamprophyres have extrusive as well as intrusive occurrences (Stewart, 1970; Larsen, 1977; Hayward and Jones, 1991; Andersen, 1997). Whereas a compositional spectrum may exist between doleritic ('basaltic') rocks and ultramafic lamprophyres in the Border Zone, with gradation via feldspathic lamprophyres, a compositional discontinuity appears to exist between basaltic rocks and ultramafic lamprophyres in the batholith-axial zone.

The basic magmas typically ascended as dykes (Fig. 2) and the early lavas probably erupted through fissure volcanoes. In contrasting style, the salic magmas more generally arose by stoping or cauldron subsidence to form central-type complexes in which syenitic rocks predominate. Some 12 such complexes are known. Despite the wide age-range, the alkaline plutons show remarkable mutual affinity in terms of internal structure, petrography and geochemistry and constitute a distinctive igneous province.

From the preservation of sediments and lavas and the fact that many intrusions are drusy and/or amygdaloidal, current erosion levels are inferred to be generally 2–3 km below contemporary surface levels and unlikely to exceed 5 km. Since the magmas attained such high crustal levels many are likely to have had surface expression (Emeleus and Upton, 1976; Upton et al., 1990).

The spelling of Greenlandic names was revised in the 1980s and new spellings are used for place names throughout the present article. However, names subject to the Rules of Stratigraphic Nomenclature are unaffected, including names of most of the igneous complexes.

2. Supracrustal rocks

Poulsen (1964) coined the name Eriksfjord Formation for the ~3.4-km-thick sequence of sedimentary rocks and lavas seen in their fullest development between Nordre Sermilik and Tunulliarfik fjords where the formation is bounded by an ENE- to WSW-trending fault. The succession comprises some 1.8 km of clastic sediments and approximately 1.6 km of volcanic rocks and there appears to have been an intimate relationship between faulting, sedimentation and volcanism. The sedimentary rocks in the Eriksfjord Formation are mainly fluviatile and aeolian sandstones (Poulsen, 1964; Clemmensen, 1988; Tirsgaard and Øxenvad, 1989).

Poulsen (1964) divided the formation into six members. The lowest, Mâjût Member, comprises sandstones and conglomerates. The latter, together with younger conglomerates intercalated with the earliest lava units in the overlying Mussartût Member, indicate that the adjoining highlands experienced successive relative uplifts resulting from movements on faults bounding the subsiding basin. The Mussartût Member also contains the extrusive and hypabyssal intrusive alkaline ultramafic and carbonatitic rocks of the Qassiarsuk Complex (Stewart, 1970; Andersen,

1997). A carbonated ultramafic breccia underlying the lowest basaltic lava may be correlative with the Qassiarsuk Complex. The third (Naujarssuit) member consists wholly of arenites while the fourth, Ulukasik Member, contains lavas erupted in a sandy desert environment. Well-preserved pahoehoe surfaces on some of the lavas demonstrate that erosion between eruptions was minimal and that magma productivity at this stage was high. The sandstones of the fifth (Nunasarnaq) member are overlain by the (Ilímaussaq) member which is made up almost entirely of lavas, namely basalts passing up into hawaiites, trachybasalts, trachyandesites, trachytes and phonolites (Stewart, 1964). Thickening, and an increase in the number of the Eriksfjord Formation lava flows towards the north and east, suggests that eruption sites lay towards Nordre Sermilik and eastern Tunulliarfik fjords (Larsen, 1977).

Outliers of the Eriksfjord Formation occur to the E and ENE of the type section where they are intruded by components of the Igaliko Nepheline Syenite Complex. At Narsarsuaq (close to South Qôroq), a succession dominated by basalt lavas (>100 m thick) but also containing sandstones and a few thin carbonate-rich tuffs is cut by the South Qôroq and North Qôroq syenite complexes. The Motzfeldt syenite complex intrudes sedimentary and volcanic rocks correlated with Mâjût and Mussartût Members of the Eriksfjord Formation.

Scattered evidence in the form of neptunean dykes in the Ketilidian terrane, xenoliths in the Gardar intrusions and clasts from glacial moraines demonstrate that the Eriksfjord Formation was originally much more widespread than is indicated by the surviving outliers and was not restricted to the rifted zone extending ENE from Tugtutôq towards the Inland Ice (Watt, 1966; Kalsbeek et al., 1990). In view of the absence of unconformities, the entire Eriksfjord Formation may have accumulated within a few million years (Larsen, 1977; Piper et al., 1999). The lava succession exhibits multiple reversals of the magnetic field (Piper, 1977; Thomas and Piper, 1992) and the Ulukasik lavas appear to have erupted during a brief part of a polarity inversion (Thomas and Piper, 1992).

The age of the Eriksfjord Formation remains contentious and recent work has tended to exacerbate the problem. Dates of $\sim$1170–1200 Ma have been obtained from lavas within the Ulukasik Member (Paslick et al., 1993), within error of dates from Mussartût lavas (Andersen, 1997). However, these comparatively young dates are at variance with Rb–Sr radiometric determinations on the oldest Igaliko Syenite Complexes, namely North Qôroq and Motzfeldt (Table 2). These complexes, both of which contain Eriksfjord Formation xenoliths and intrude Eriksfjord Formation lavas and sediments, are dated at 1268 ± 60 and 1282 ± 30 Ma, respectively. Using a MacIntyre 'Model 3' regression (Ludwig, 2001) provides slightly younger ages (1244 ± 17 and 1265 ± 16 Ma, respectively) that overlap within 2σ with the Rb–Sr date of the Qassiarsuk Complex (Andersen, 1997). On this basis, it could be maintained that the base of the Eriksfjord succession is synchronous with the older Igaliko centres and further radiometric analysis may provide younger dates closer to $\sim$1.2 Ga for those plutons (see, e.g. Finch et al., 2001a). However, a U–Pb dating of a Motzfeldt pegmatite (Table 1) at 1275.3 ± 1.1 Ma lends support to the older dates for Motzfeldt and North Qôroq as determined using the Rb–Sr method (Table 2). Clearly, the data remain subject to different interpretations. For example, Piper et al. (1999) regarded the dates from the Ulukasik lavas as having been wholly reset by younger Gardar magmatism. They preferred the previously held opinion of a 1350–1300 Ma age on the basis of palaeomagnetic evidence, since the multiple magnetic reversals present within the formation are better accommodated within the mixed polarities known, from Laurentian data, to have existed in the interval 1400–1300 Ma.

The estimated 2–5 km of eroded 'cover' is likely to have been composed of continental clastic sediments and volcanic products from younger Eriksfjord Formation rocks as well as from post-Eriksfjord Formation dykes and central volcanoes.

3. Dykes

The dolerite dykes, which have distinctive brown weathering, have accordingly been widely referred to as 'brown dykes' or 'BDs'. The earliest brown dykes (BD_0s) form an important WNW- to ESE-trending swarm across the Province. The BD_0 swarm is >90 km across and can be traced for over 200 km: typical

dykes have widths exceeding 60 m and some, whose wall rocks are extensively rheomorphosed, may have acted as conduits to surface volcanism. The observation that many are sheared parallel to their length may be due to periodic adjustments on the E–W transcurrent faults (Bridgwater and Coe, 1969). U–Pb baddeleyite dating on three BD_0 dykes, from the Qaqortoq, Kangerluarsuk and Tugtutôq regions, gave similar dates of 1284 ± 3, 1280 ± 2 and 1279 ± 1.3 Ma, respectively, (Table 1).

In the Isortoq region, close to the northern margin of the Julianehåb batholith, the BD_0s were followed by a few N–S dykes (Bridgwater and Coe, 1969). A third generation of intrusions constitutes a group of ENE-trending troctolitic dykes and plugs with widths up to ~ 100 m. A period of faulting intervened between these and a still younger suite of ENE dykes. The mid-Gardar dyke swarm through the Isortoq area is part of a major regional swarm extending from the Nunarssuit Complex to the Inland Ice. It contains several generations of basic dykes (dolerites and gabbros) together with microsyenites and culminates in four broad (up to 500 m) composite 'giant dykes' with granitic or syenitic centres enclosed in syenogabbroic margins. These giant dykes extend WSW to Bangs Havn and Eqaloqarfia (Pulvertaft, 1965) where they are cut by the syenites and granites of the Nunarssuit Complex.

Whereas in the area between the Nunarssuit Complex and Isortoq area a complex array of basic and intermediate dykes was intruded between the BD_0 event and the giant dykes, this is not true of the batholith-axial zone. In the latter, intrusive activity was essentially absent during the approximately 100-Ma interval between the BD_0 intrusions and the emplacement of the 1184 ± 5 Ma Tugtutôq Older Giant Dyke (Table 1). The Older Giant Dyke event was a precursor to the much larger, dominantly troctolitic, 1163–1166 Ma Younger Giant Dyke which expanded into a lopolithic body at the unconformity between the Julianehåb granite and the overlying Eriksfjord supracrustal formation at Narsaq (Upton and Thomas, 1980). During subsequent prolonged cooling and reduction of extensional stress, a sequence of ENE dykes was intruded along the batholith-axial zone, whose individual widths tended to decrease with time while their degree of differentiation increased culminating in trachytes, phonolites and comenditic rhyolites.

The entire dyke swarm, including the giant dyke(s), may have underlain an elongate fissure-fed volcanic system analogous to some volcanic lineaments in Ethiopia (Barbieri and Varet, 1970) and on the Arabian side of the Red Sea (Camp et al., 1989). Since these possible Phanerozoic analogues developed in zones of rapid crustal extension, it is reasonable to infer the same for the batholith-axial lineament. Giant dykes do not occur in the Border Zone in which the BD_0 dykes were followed by two more generations of 'brown dykes' (Berthelsen, 1962; Berthelsen and Henriksen, 1975) with successively more northeasterly trends. Other basic and alkaline salic dykes also occur in this area.

The concentration of late Gardar dykes within the Nunarssuit–Isortoq and the batholith-axial zones may imply that these zones marked the sites of subsiding rift basins with higher ground to the north and south and with the two rift basins separated by a horst.

4. Giant dykes

Although there is every gradation from small dykes to 'giant' dykes, the latter, with widths of 200–800 m, have some idiosyncratic features that justify separate treatment. Whereas giant dykes with a generalised ENE–WNW trend are common to both the Nunarssuit–Isortoq and the batholith-axial zones, they attained their fullest development in the latter. Some smaller sections of analogous giant dykes occur north of Tugtutôq, on the island of Tullerunnat Killiit and in the South Qôroq Complex. Giant dyke occurrences north of Motzfeldt are regarded as extensions of those on Tugtutôq. The Tugtutôq Giant Dykes (and, by inference, the others) were intruded relatively late in the Gardar (Table 1). These big dykes can be composite, either in whole or in part (Fig. 3). The great widths imply the operation of intense stresses acting on mechanically weak crust while the lack of internal chilled contacts implies rapid crustal opening to accommodate the magmas. Although brittle failure attending intrusion is demonstrable for the small Gardar dykes, Bridgwater and Coe (1969) called attention to the problem presented by the 'pinch and swell' features exhibited by the Isortoq giant dykes (Fig. 3). The contacts of the Tugtutôq Younger Giant Dyke (Fig. 3) similarly present nonparallelism with

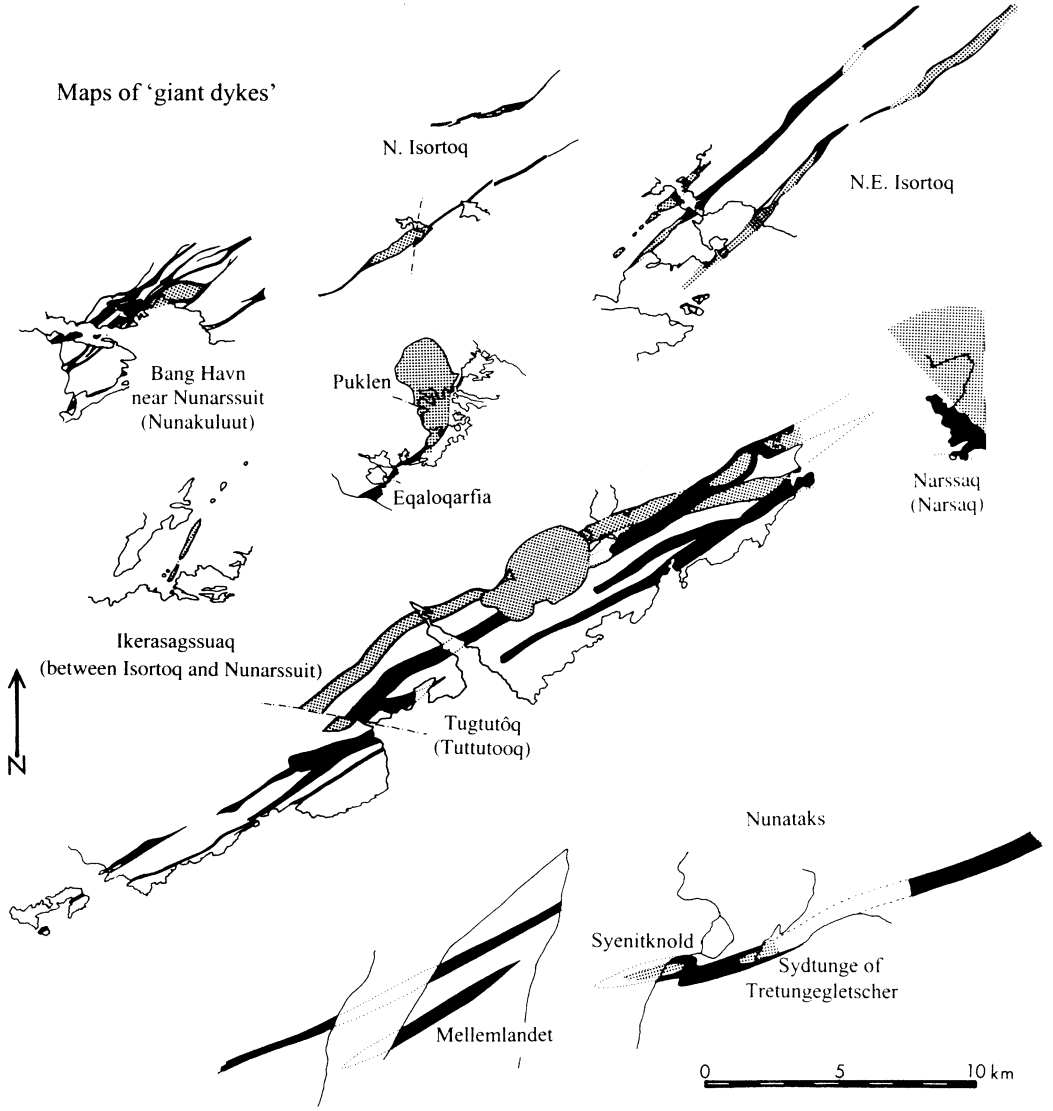


Fig. 3. A composite map showing the outcrop forms of some of the Gardar giant dykes. Basic portions indicated in black; salic components shown in stipple.

widening and narrowing in both horizontal and vertical senses. Accordingly, intrusion models involving simple dilation of the country rock granites provide an inadequate explanation. Fig. 3 shows the various outcrop forms presented by giant dykes in various parts of the Nunarssuit, Isortoq, Tugtutôq and Nunataq regions to the north and northeast of Motzfeldt.

Two of the Tugtutôq Giant Dyke branches have abrupt western terminations, aligned NNW–SSE, i.e. almost perpendicular to the overall trend of the intrusion (Fig. 3). It is suggested that the dyke magma, fed in with a west-south-westerly vector, encountered NNW-trending fractures that acted as small-scale transform faults as the dyke inflated.

Bridgwater and Coe (1969) stressed the importance of stoping in the emplacement of the syenites in the Isortoq giant dykes. Stoping undoubtedly played a part in the emplacement of the syenitic facies of the

giant dykes, where low density benmoreitic to trachytic melts were involved. However, there is no incontrovertible evidence that the basaltic to hawaiitic magmas, which formed the initial gabbroic or syenogabbroic phases of the giant dykes, were ever emplaced by stoping. Furthermore, density estimates of melts and granitoid host rocks make this mechanism highly improbable.

Where composite, the giant dykes have mafic margins and more salic centres that are precisely bilaterally symmetrical. The lack of chilled contacts between these implies that the mafic and salic magmas were essentially contemporaneously available. Whereas the Tugtutôq Older Giant Dyke (Upton et al., 1985) is composite along its entire outcrop length (Fig. 3), some of the giant dykes are only intermittently composite with localised expansions around central 'pods' of more fractionated material. The latter are granitic (Bangs Havn Complex), quartz syenitic or ferro-syenogabbroic (Tugtutôq Younger Giant Dyke) or nepheline syenitic (Syenitknold, north of Motzfeldt). Despite the observation that the more basic marginal facies of these large dykes chilled to fine-grained selvedges against the host granites, implying a significant temperature difference, the latter must, nonetheless, have been hot enough to permit some ductile deformation. That the wall rock granites can have sustained ductile deformation to accommodate the localised giant dyke inflations implies relatively high temperatures and thus exceptional heat flow. Intrusion of the giant dykes may also have involved lateral movement on wall rocks other than simple dilatation. The intersection of the Tugtutôq Younger and Older Giant Dykes, for example, indicates some left-lateral displacement of the wall rocks (Fig. 3). The structural problems associated with the giant dyke phenomena are numerous and clearly require further attention. Comparable examples elsewhere in the world are few although the somewhat older, partly composite Proterozoic 'giant dykes' in central Sweden (Kroköström, 1936; Lopez-Montano, 1986) may provide analogues.

5. Central complexes

These typically comprise composite stocks and annular intrusions (Upton, 1974; Emeleus and Upton, 1976). Syenite and nepheline syenite are the principal components, although alkali granite, gabbro and syenogabbro can also be present. Ring dykes of syenitic rocks occur in Motzfeldt (Emeleus and Harry, 1970), South Qôroq (Stephenson, 1972) and Tugtutôq Central Complex (Upton, 1962) while gabbroic or syenogabbroic ring dykes are present at Igdlerfigssalik and Motzfeldt (Emeleus and Harry, 1970) and Kûngnât (Upton, 1960). Klokken (Parsons, 1979) is unique in having an outer sheath of gabbroic rocks surrounding a younger syenite core. In this respect, Klokken closely resembles the composite sectors of the composite giant dykes but, rather than having a lensoid or tabular form as in the latter, the geometric relationship at Klokken is that of concentric cylinders. The intrusive mechanism at Klokken remains enigmatic. In common with other centres, deformation of country rocks is generally minimal and the alkalic plutons were principally emplaced by 'cauldron subsidence' or piecemeal stoping of the country rocks. Additionally, at Klokken and Ilímaussaq, the syenitic magmas were able to break through their own earlier crystallised and somewhat denser roofing facies by stoping (Upton et al., 1996 and references therein).

Layered cumulates, coarse grain-sizes and strong in situ crystal fractionation characterize most of the larger Gardar intrusions and have been related to unusually low viscosity in fluorine-rich melts (Upton et al., 1996). In some complexes (South Qôroq and Kûngnât), successive intrusions were increasingly primitive implying derivation from compositionally stratified magma chambers (Stephenson and Upton, 1982). Intrusive sequences in some multiple dykes provide confirmatory evidence that stratified magmas existed.

Some Gardar complexes (e.g. Motzfeldt, Qôroq and Igdlerfigssalik) are consistently silica-undersaturated, whereas others (e.g. Nunarssuit) are wholly oversaturated. Irrespective of age, the silica-oversaturated complexes occur along the southern and western sides of the province whereas the undersaturated complexes occur to the north and east. However, in the vicinity of the Narsaq and Ilímaussaq complexes, silica saturation appears to have oscillated with time across the alkali feldspar divide in 'the residua system'. Nepheline syenites of the Tugtutôq Older Giant Dyke were followed by silica-oversaturated syenites in the Younger Giant Dyke and in the Dyrnæs–

Narsaq Complex. These were succeeded by the undersaturated 'augite syenite' intrusion of the Ilímaussaq Complex. Quartz syenites and alkali granites came next, followed by the silica-undersaturated agpaitic suite, emplaced as the youngest component at Ilímaussaq. If, following Foland et al. (1993) and Stevenson et al. (1997), the degree of silica saturation was controlled by the amount of crustal contamination of parental magmas, the variation in degrees of assimilation across the province may have implications for lateral geothermal gradients in the Ketilidian crust.

6. Anorthosite xenoliths and plagioclase megacrysts

Anorthositic xenoliths (up to 100 m across) and plagioclase megacrysts (up to 1 m) are common in the basic and intermediate intrusions but are extremely rare in the syenites and other evolved hosts. The coarse-grained anorthosites, typically composed of intermediate plagioclase (An_{65-55}), have been described in detail (Bridgwater and Harry, 1968). Bridgwater and Harry recognised wide textural variation within the xenoliths ranging from massive granular anorthosites, commonly displaying evidence of crushing, to those with unmodified cumulate textures. Cataclasis in the granular xenoliths has been ascribed to pene-contemporaneous faulting (Bridgwater, 1967). Granular anorthosite xenoliths in several localities themselves contain texturally distinct anorthosite clasts, providing further evidence for evolutionary complexity in the parent body and the likelihood that diapirism was involved in its emplacement.

The principal intercumulus component of the cumulate textured xenoliths is olivine, accompanied by subordinate augite, FeTi oxides, apatite and biotite (Upton, 1964, 1974). Excellent feldspar lamination in the cumulate xenoliths is commonly associated with layering defined either texturally, or by modal variation due to different proportions of cumulus plagioclase and intercumulus ferromagnesian minerals or to both factors. The bulk of the anorthosite xenoliths have alkaline affinities suggesting that they are cognate with their hosts (Upton, 1996). However, xenoliths from some intrusions in the north of the province, cutting the Archaean foreland (Fig. 1), contain olivine and subordinate orthopyroxene. Megacrysts of olivine and orthopyroxene also occur (Bridgwater and Harry, 1968).

The occurrence of the feldspathic inclusions over the entire province led to the conclusion that they were derived from a subsurface anorthositic complex occupying an area of $\sim 20{,}000$ km^2 (Bridgwater, 1967; Bridgwater and Harry, 1968). The xenoliths and megacrysts occur in the gabbros and dolerites but are particularly associated with the dykes of intermediate composition (typically mugearitic) that were accordingly referred to as 'big feldspar dykes'.

According to Bridgwater and Harry, plagioclase megacrysts compose up to 75% of the total feldspathic inclusions in the Gardar intrusions. In many instances, megacrysts occur together with xenoliths but the proportions can vary widely. In some (e.g. the Klokken gabbro), great masses of texturally contrasted anorthosites occur unaccompanied by megacrysts. However, while some megacrysts are almost certainly products of disaggregated xenoliths, Bridgwater and Harry (1968) noted characteristics of the megacrysts that they could match in neither the granular nor the laminated anorthosites. They therefore appear to pose a related but separate problem. Some megacrysts exhibit euhedral morphologies (Bridgwater and Harry, 1968; Halama et al., 2002) and may be regarded as high-pressure phenocrysts.

A detailed study of zonal relationships in plagioclase megacrysts from 'big feldspar dykes' in the Isortoq region indicated early crystallisation at 10–12 kbar (30–35 km depth) followed by relatively rapid ascent into the upper crust where further crystallisation ensued (Halama et al., 2002). These data are in harmony with the concept that anorthosite crystallisation commenced close to the crust–mantle boundary. Nonetheless, some of the xenoliths possess orthocumulate textures indicative of faster cooling regimes that would be expected in the deep crust and a mid-crustal environment for their crystallisation (10–15 km below the former surface?) is suggested.

7. Tectonics

The numerous faults affecting the Gardar Province form ENE- and E-trending and NNW- to NNE-trending groups. The first group comprises left-lateral

wrench faults while the second group involves mainly right-lateral displacements. These, apparently conjugate, faults operated throughout the Gardar becoming virtually inactive by ~ 1160 Ma when some of the younger alkaline complexes (e.g. Ilímaussaq) were intruded. Berthelsen and Henriksen (1975) proposed

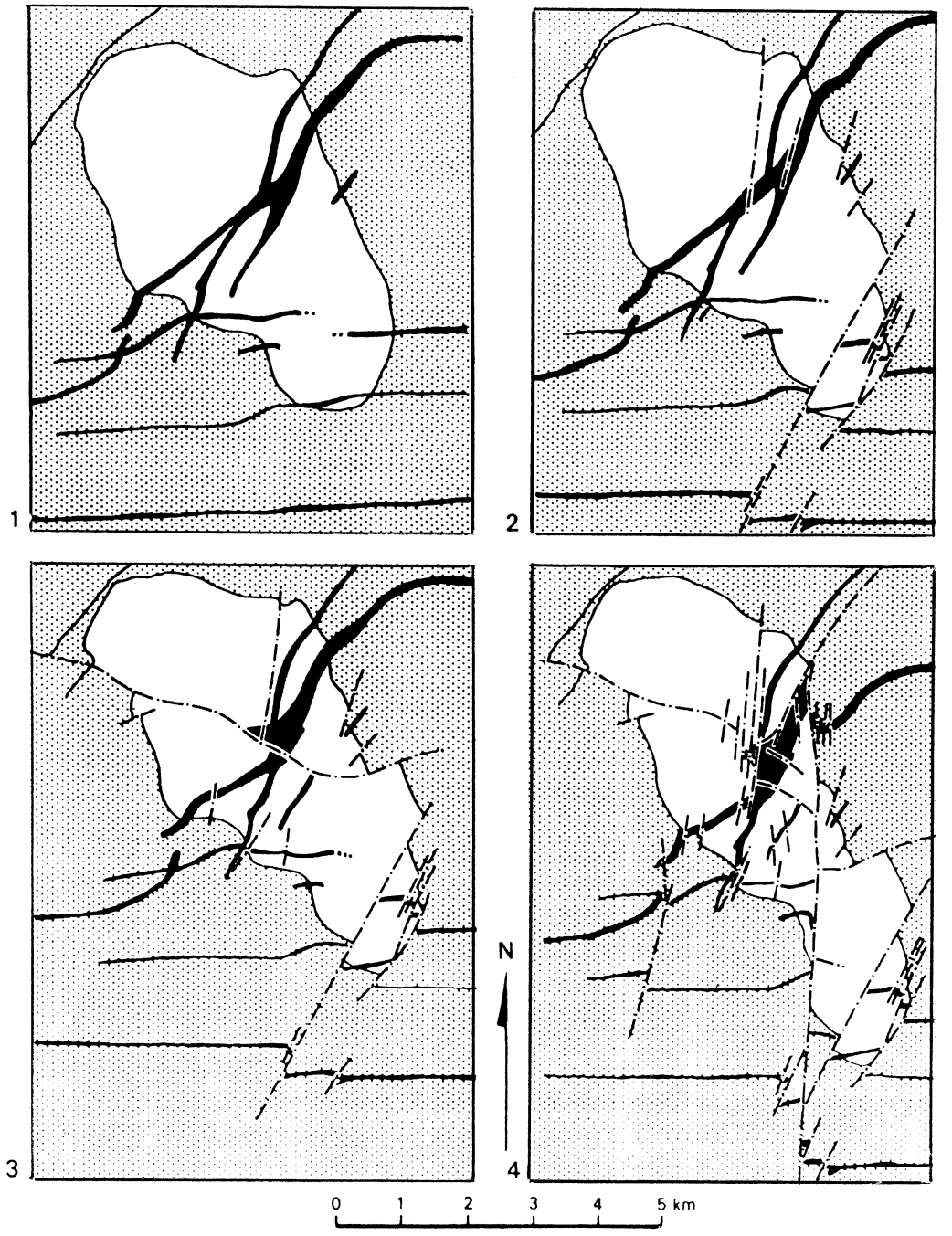


Fig. 4. The Grønnedal–Íka Complex; 1–4: stages in the development of faulting in the history of the complex (after Emeleus and Upton, 1976).

that dyke intrusion and faulting developed in a single stress regime whose minimum axis changed with time from ~E–W to NW–SE. The ENE–WSW faults, which developed at intervals some 10 km apart, collectively gave rise to a total left-lateral displacement of 15–20 km. These faults segmented the late Gardar batholith-axial zone (rift) in a manner possibly analogous to that described for the North American Midcontinental Rift System (Green, 1992 and references therein).

Some of the ENE–WSW left-lateral faults can be traced long distances and one at least (the Laksenæs Fault, named after a locality close to Grønnedal) had pre-Gardar origins (Henriksen, 1960; Berthelsen and Noe-Nygaard, 1965). The Laksenæs Fault and other fault systems transect the Grønnedal–Íka Complex in the northwestern part of the province (Fig. 4) where intrusion and fault movement tended to alternate (Emeleus, 1964). A similar relationship between faulting and dyke intrusion was recorded by Berthelsen and Henriksen (1975) from the adjoining peninsula south of Ivigtut. Further complex intercalated relationships between dyke intrusion and faulting were deduced from the Qaarsuarsuk area (Watt, 1968).

The Gardar plutons situated at intersections of regional dyke swarms and left-lateral (~E–W) faults tend to be elliptical in plan with long axes directed NW–SE (e.g. Grønnedal–Íka, Igaliko, Ilímaussaq and, to a lesser degree, Dyrnæs–Narsaq, Nunarssuit and Puklen). In contrast, those remote from such faults (e.g. Kûngnât, Tugtutôq Central Complex and Klokken) are more nearly circular (Stephenson, 1976). Stephenson accounted for this behaviour with a simple shear model which postulated that the elliptical plutons experienced ductile deformation by pene-contemporaneous faulting while still at high temperatures. The plan form of these intrusions, believed to have initially been circular, subsequently approximated to that of a simple strain ellipse.

The left-lateral WNW-trending faults may have provided channel ways for rising magmas. A section of giant dyke (~500 m wide) on Tullerunnat Killiit appears to have been intruded along one of these faults. WNW-trending contacts of sections of the Younger Giant Dyke on Tugtutôq and also at Narsaq may have been controlled by pre-existing faults.

The WNW–ESE fault transgressing the Dyrnæs–Narssaq Complex has an apparent left-lateral shift of ~2 km and a significant down-throw to the north (Hamilton, 1964). Although it may have controlled the location and form of the subsequent Ilímaussaq Complex (Ussing, 1912; Ferguson, 1964), the latter shows no brittle fracture or displacement. However, fine flaser structures within the earliest unit of the Ilímaussaq Complex together with a zone of haematite staining across the northern part of the complex suggest that minor strain and fracturing persisted until late Gardar times. The principal movements on this fault are thus essentially age-bracketed by the Younger Giant Dyke at ~1166 Ma (1167 ± 3 Ma U–Pb baddeleyite date from this study, Table 1; and 1165.7 ± 1.2 Ma; updated U–Pb baddeleyite date of Buchan et al., 2002 (M.A. Hamilton personal communication) and the Ilímaussaq Complex at ~1160 ± 5 Ma (unpublished U–Pb baddeleyite date reported in Waight et al., 2002). A pattern of curved joints with a radius of ca. 30 km, centred around Ilímaussaq, has been discerned from satellite images (Amour-Brown et al., 1983). While these crudely concentric fractures are considered to long predate the Ilímaussaq intrusion (Kalsbeek et al., 1990), their significance remains enigmatic.

8. Ultramafic lamprophyres and carbonatites

Lamprophyres and carbonatites are minor components of the Province. Apart from the wide spread of lamprophyre varieties (e.g. vogesites, spessartites and camptonites) within the dyke swarm in the Grønnedal–Íka area (Emeleus, 1964), ultramafic lamprophyres (aillikites and alnöites; Rock, 1986) have such a close relationship to the carbonatites as to suggest related geneses. Both lithologies occur in the batholith-axial zone. Carbonatites and ultramafic lamprophyres also occur in the Border Zone whereas they are notably absent from the Nunarssuit–Isortoq Zone.

Aillikites occur, mainly as minor intrusions, in the more east-north-easterly parts of the batholith-axial zone. The eroded volcanic centre at Qassiarsuk is composed of a comagmatic assemblage of carbonatites and feldspar-deficient silicate rocks (monchiquites, mica-peridotites, melilitites, alnöites and mica pyroxenites: Stewart, 1970; Hayward and Jones, 1991; Andersen, 1997) related to the ultramafic lamprophyres. The volatile-rich aillikite magmas ascended

sufficiently rapidly to have retained primitive compositions and, occasionally, to carry mantle xenoliths (Upton, 1990; Finch et al., 2001b).

'Mid-Gardar' ultramafic lamprophyre dykes in the Border Zone preceded the dolerite dyke swarms, whereas the majority are younger than the nepheline syenites and the carbonatite (sövite-ferrocarbonatite) plug at Grønnedal–Íka, a lamprophyre xenolith, occurs in one of the syenite units. Pearce and Leng (1996) observed that the composition of the Grønnedal–Íka carbonatites closely matches those of the Igaliko area in the batholith-axial zone. They concluded, from a C and O isotopic study, that they are related to the ultramafic lamprophyre magmas rather than to carbonated phonolite magmas as had hitherto been maintained (Emeleus, 1964; Pearce, 1988; Bedford, 1989; Hamilton et al., 1989). Although the hiatus between the carbonatites and the carbonated ultramafic lamprophyric association may be due to a liquid immiscibility relationship between the two groups (Pearce and Leng, 1996; Andersen, 1997; Pearce et al., 1997), the compositional gap is small and the possibility of a continuum cannot be wholly discounted (Upton and Fitton, 1985). The entire suite may represent products of variable degrees of mantle melting (Upton and Fitton, 1985; Pearce and Leng, 1996). Andersen (1997) concluded that the Qassiarsuk carbonatite and its associated silicate rocks were derived from a LILE-and LREE-depleted, mantle source ($\varepsilon_{Nd}>4$, $\varepsilon_{Sr}<13$, $\mu \leq 7.4$) that may have been metasomatically enriched in CO_2 and other incompatible elements prior to melting.

Differential metasomatism of the sub-Gardar lithospheric mantle (Upton and Emeleus, 1987), brought about by incremental addition of carbonatitic small-fraction melts, may have had particular relevance to the genesis of the Ivigtut intrusion. The Ivigtut granite hosted the fluoride-rich pegmatite body from which cryolite was mined. Prior to extraction, this complex pegmatite (Pauly and Bailey, 1999) was not only rich in fluorides but also in carbonate (principally siderite) and sulphides. From an isotopic and geochemical study, Goodenough et al. (2000) proposed that the granite originated through assimilation-fractional crystallisation processes acting on mantle-derived magma, with post-magmatic involvement of CO_3^{2-} and F^--rich metasomatic fluids, initially derived from F-rich carbonatites.

9. Nature and genesis of the subsurface anorthosite complex

There has been extended debate concerning whether the Proterozoic massif-type anorthosites were derived from mantle-derived basaltic (normally tholeiitic) high-Al magmas or from aluminous crustal sources (Ashwal, 1993; Longhi et al., 1999; Bédard, 2001 and references therein). Bédard addressed the contentious matter as to whether or not anorthosites and associated mafic cumulates are cogenetic. In the Gardar Province, however, there appear no valid reasons for doubting the genetic affinity of the (alkalic) anorthosite xenoliths and the gabbroic cumulates and the relatively fast cooled hawaiitic to mugearitic dyke rocks in which they occur. Fe–Ti–P-rich ferro-syenogabbroic and ferro-syenodioritic rocks in the Gardar, which may be regarded as more alkalic analogues of the ferrodiorites, jotunites, monzodiorites, etc. of the North American associations, are interpreted as crystal-fractionation products from the basaltic magmas (Upton, 1987). The principal reasons for regarding crustal assimilation as having been insignificant in the formation of the anorthosites are (a) their generally alkalic character and (b) their low initial $^{87}Sr/^{86}Sr$ values (Patchett et al., 1976).

The Gardar complex has been inferred to be composed of plagioclase flotation cumulates, developed sequentially as a polyphase body over the whole time span of the Gardar (Bridgwater, 1967; Bridgwater and Harry, 1968). The component feldspars are predominantly sodic labradorites with the implication that they crystallised from melts with higher Na/Ca ratios than those of basalt. The anorthosite xenoliths and plagioclase megacrysts are particularly abundant in dyke rocks of hawaiitic and mugearitic composition. Iron contents reach their maxima in this part of the Gardar magma series and melt densities would have been correspondingly high.

10. Basaltic magmas and their derivatives

The gabbroic and doleritic intrusions and basic lavas crystallised from relatively evolved basaltic and hawaiitic magmas: $100 \times Mg/(Mg+Fe'')$ values of the more primitive samples rarely exceed 30–40. These magmas were transitional to mildly alkalic,

notably aluminous and with normalised trace element patterns typically unlike those of oceanic island basalts in having higher La/Nb and lower La/Ba, La/Rb and La/K (Upton and Emeleus, 1987). Their evolved state is attributed to a previous history of fractional crystallisation involving loss of olivine, spinel (± pyroxene). A long history of crustal underplating beneath both the Ketilidian batholith and the Border Zone, with generation of peridotitic cumulates, may be inferred. The ubiquitous high Al/Ca ratios of the basaltic magmas may denote derivation from mantle sources that had been largely depleted through earlier (Ketilidian) basalt extraction (Goodenough et al., 2002).

That Gardar basaltic magmas had trace element patterns consistently different from those of oceanic island basalts precludes their direct generation from the asthenospheric mantle (Upton and Emeleus, 1987; Fitton and Dunlop, 1985). Basanitic and nephelinitic magmas are regarded as products of small-fraction asthenospheric melts and their absence from the Gardar Province provides supplementary evidence that asthenospheric melting can have been only indirectly involved. Accordingly it is suggested that the Gardar basic magmas may represent melts from lithospheric mantle and that their high Al/Ca (troctolitic) signatures relate to an origin from dominantly diopside-poor peridotites residual after earlier melt extraction. Metasomatic enrichment of such refractory peridotites during lithospheric attenuation is inferred to have facilitated partial melting (Upton, 1996).

Basaltic magmas that attained shallow levels across the province from early to late in the Gardar were commonly supercooled and oversaturated with respect to plagioclase. One line of evidence for this is the occurrence of ‘snow-flake’ glomerocrysts (‘star-basalts’) within the upper members of the Eriksfjord Formation lavas (Stewart, 1964) and many of the mid- to late-Gardar doleritic dykes. It has been argued that layering in the more primitive zones of the Tugtutôq Younger Giant Dyke was due to alternating cumulus batches of (1) olivine and plagioclase ‘snow-flake’ clusters and (2) olivine alone (Upton et al., 1996). Features described in the Eqaloquarfia giant dyke (near the Nunarssuit Complex) by Pulvertaft (1965) appear similarly explicable. Further indications of plagioclase supersaturation are the ‘perpendicular feldspar rocks’ (cf. Wager and Brown, 1968) in the troctolitic gabbros of the Kûngnât Complex (Upton, 1960) as well as those of the Eqaloqarfia and Tugtutôq Giant Dykes (Pulvertaft, 1965; Upton et al., 1996). The high Al/Ca ratios and Sr contents and lack of negative Eu anomalies in these rocks confirm that the magmas had not undergone significant plagioclase fractionation.

It is concluded (albeit simplistically) that basaltic magmas arising from chambers close to the crust–mantle boundary in which extensive ferromagnesian crystal fractionation had occurred experienced one of two fates. Either they were subsequently held at depth for further fractionation (adding their feldspar increment to the developing anorthositic complex in so doing) or, in more energetic extensional episodes, penetrating this complex to arrive in a plagioclase-supersaturated ‘hyperfeldspathic’ (cf. Wiebe, 1990) state at high crustal levels. Rapid crystallisation of these supersaturated magmas yielded ‘star-basalt’ lavas and dykes whereas slower-cooling in larger intrusions produced ‘snowflake’ cumulates or perpendicular feldspar crescumulates (Wager and Brown, 1968).

The very low bulk distribution coefficients for the incompatible elements Ce, Y, Zr and Nb are such that their ratios in basaltic, ultramafic lamprophyric and carbonatitic rocks should reflect those of their mantle sources. The wide range of Ce/Y vs. Zr/Nb for the basic, ultrabasic and carbonatitic rocks of the Gardar Province implies that these sources were geochemically heterogeneous (Fig. 5). The carbonatite-aillikite suite, with high Ce/Y and low Zr/Nb, is inferred to have originated from enriched sources and accords with the hypothesis that the latter constituted metasomite facies in the deep lithosphere. Glimmerite xenoliths accompanying (altered) peridotite xenoliths in one aillikite intrusion (Upton, 1990) may be tangible evidence for the presence of such metasomites.

Whereas the lamprophyre/carbonatite association is characterised by high Ce/Y and low Zr/Nb, the basaltic rocks of the Eriksfjord Formation, the NE and ENE dolerite dykes of the Border and Nunarssuit–Isortoq Zones and the WNE–ESE BD_0 dykes have notably lower Ce/Y and higher Zr/Nb (Fig. 5). This suggests that their mantle sources were poor in LREE and Nb relative to Y and Zr in comparison with the basaltic and hawaiitic dykes of the batholith-axial zone. These latter tend to have Ce/Y and Zr/Nb values

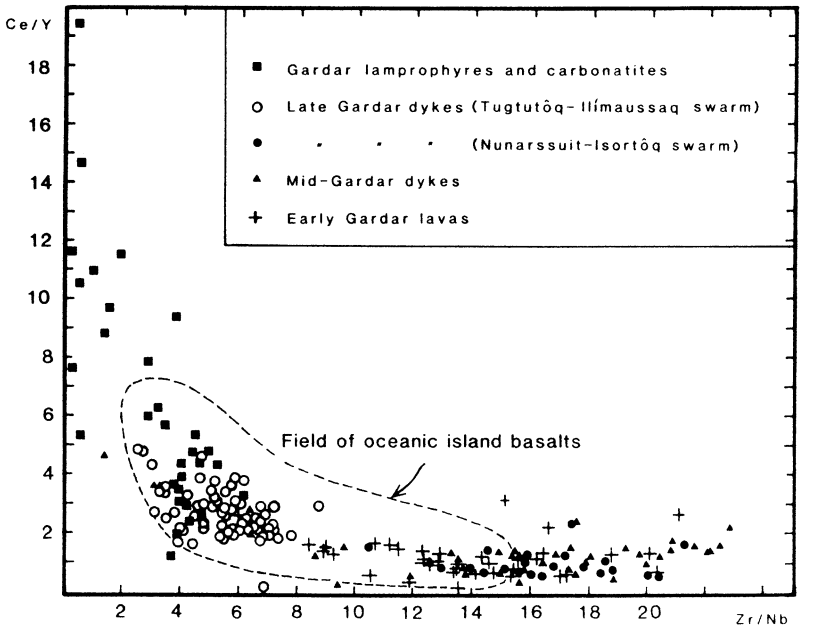


Fig. 5. Zr/Nb vs. Ce/Y diagram for Gardar basic lavas, doleritic dykes, gabbroic chilled margins, ultramafic lamprophyres and carbonatites (based on unpublished data, B.G.J. Upton).

intermediate between the more extreme situations considered above. LREE enrichment is distinctly more pronounced in the batholith-axial basic dykes (with La/Yb values of 11–19) than in the preceeding basic lavas of the Eriksfjord Formation (La/Yb, ranging from 3 to 9; data from Martin, 1985 and Upton, unpublished). From this it is deduced that metasomatic enrichment of the mantle sources of the batholith-axial zone in LREE and Nb was significantly enhanced in the interval separating eruption of the Eriksfjord Formation lavas and the intrusion of the Older Giant Dyke which marked the start of the Late Gardar magmatism.

11. Isotopic evidence

Isotopic data on the Gardar igneous rocks remain sparse. The first major isotope study was that of Blaxland et al. (1978) in which initial $^{87}Sr/^{86}Sr$ ratios for Gardar intrusions were found to be broadly divisible into three groups. Most, including the majority of the syenites and the basic rocks, have low ratios of 0.702–0.704. Ratios from 0.704 to 0.707 characterize certain intrusions, particularly those of oversaturated syenites and granites. Higher ratios (>0.707) were found only in two, conspicuously mineralised and metasomatised intrusions, Ivigtut and Ilímaussaq (Stevenson et al., 1997; Goodenough et al., 2000).

Subsequent Sr and Nd isotopic work has tended to support the broad division between Gardar intrusions formed from essentially uncontaminated mantle-derived magmas and those in which there has been a significant amount of crustal contamination (see Fig. 6). Various workers (e.g. Pearce and Leng, 1996; Pearce et al., 1997; Andersen, 1997; Goodenough et al., 2002) have shown that the most primitive Gardar magmas, including lamprophyres, dolerites and carbonatites, have initial $^{86}Sr/^{87}Sr$ ratios of around 0.703 and initial ε_{Nd} values of +2 to +6. In general, the lamprophyres have ε_{Nd} towards the upper end of that range. These ratios are consistent with derivation of all the parental Gardar magmas from lithospheric mantle that was moderately enriched relative to MORB. Many of the more evolved Gardar rocks, including granitic and syenitic compositions, have higher $^{87}Sr/^{86}Sr_i$ ratios and lower initial ε_{Nd} values than the less differentiated compositions. These variations are taken to represent involvement of crustal material in the genesis of the magmas.

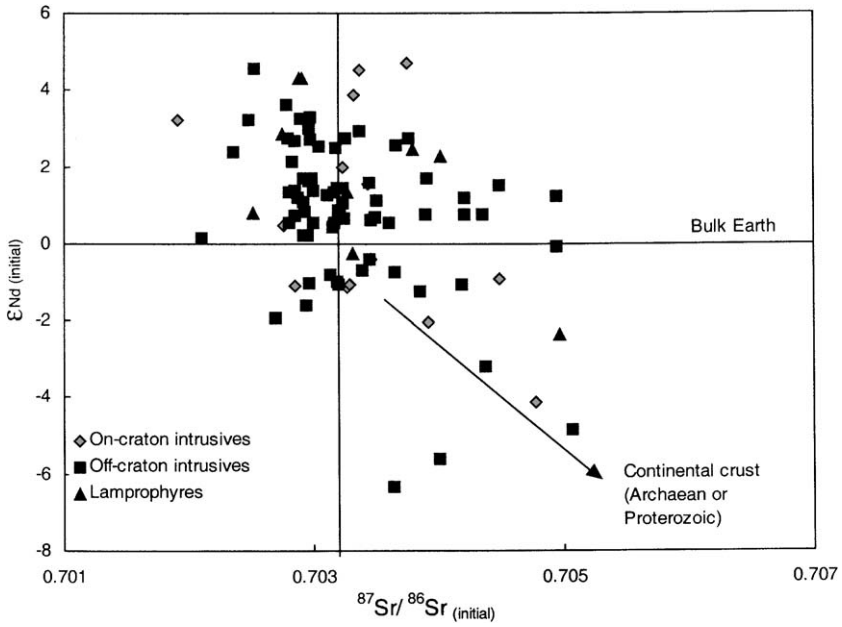


Fig. 6. Initial ε_{Nd} and $^{87}Sr/^{86}Sr$ data for on- and off-craton Gardar intrusions. On-craton intrusives include Grønnedal–Íka syenites, Kûngnât syenites and gabbros, Ivigtut granite and doleritic dykes. Off-craton intrusives include the Tugtutôq Giant Dykes and Central Complex, the Qassiarsuk Complex and late Gardar doleritic dykes. Diagram after Goodenough, 1997.

Taylor and Upton (1993) investigated the Pb isotopic compositions of rocks from the Kûngnât Complex within the Archaean Border Zone and from the Tugtutôq Younger Giant Dyke in the batholith-axial zone and showed that there are significant differences between these two complexes, as well as marked Pb isotope variations within each. The differences between the two complexes were taken to reflect the differences between the two types of crustal environments, whereas the intra-complex var-

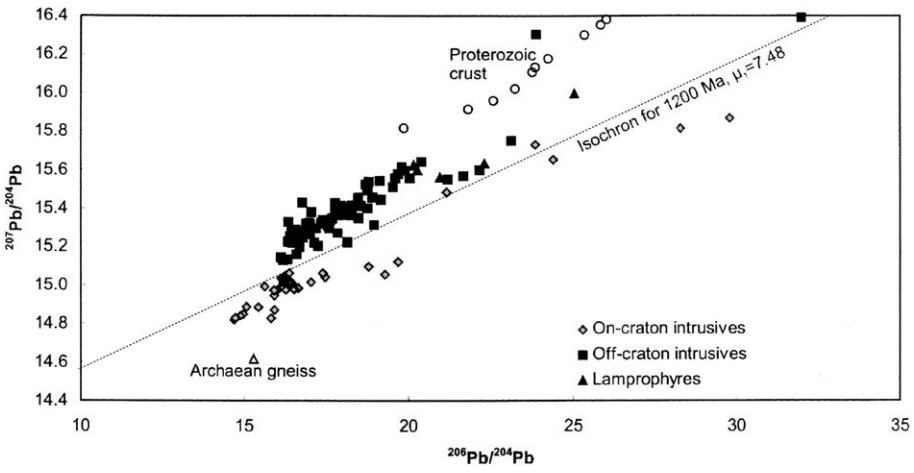


Fig. 7. Present-day Pb isotope correlation diagram for all available Gardar data, showing that ‘on-craton’ and ‘off-craton’ intrusives show trends towards the Pb isotope compositions of the types of country rocks into which they were emplaced. On- and off-craton intrusions as for Fig. 6.

iations were ascribed to varying degrees of crustal contamination. Subsequently, Goodenough (1997) collated all available Nd, Sr and Pb data for the Gardar intrusions (Figs. 6 and 7) and demonstrated that, although Nd and Sr ranges are similar across the craton boundary, there is a marked variation in the Pb isotope data for more evolved Gardar rocks which are dependent on the nature of the country rock. Pb isotopes appear to provide an excellent test for crustal contamination in the Gardar rocks and it is clear that many of the more evolved magmas assimilated substantial amounts of crust at relatively high levels. On the basis of a small sample set, the lamprophyres appear to have Pb isotopic compositions substantially different from those of other Gardar basic rocks, providing evidence for their derivation from a different source.

The isotopic evidence suggests that the Gardar basaltic magmas were derived from broadly similar, although heterogeneous, mantle sources across the province with no systematic variation between those intruded into the Archaean craton and those in the Ketilidian batholith (Goodenough et al., 2002). The basaltic parental magmas appear to have arisen from lithospheric mantle that was mildly enriched relative to MORB, with initial values of $^{87}Sr/^{86}Sr \sim 0.703$ and $\varepsilon_{Nd} \sim +3$. The lamprophyres may have been derived from melting of volatile-rich metasomites in the lithospheric mantle, formed from carbonatitic partial melts that arose from the asthenosphere (Goodenough et al., 2002).

12. Discussion

12.1. Volumetric considerations

While a quantitative assessment is not possible, the volume of the salic magmas from which the Gardar syenitic intrusions crystallised was very large. The largest of the Gardar syenitic complexes are those at Nunarssuit and Igaliko. The Nunarssuit Complex covers an area of some 560 km^2 while the coalescing Igaliko complexes collectively crop out over 250 km^2 and represent one of the world's largest bodies of nepheline syenite. The recent discovery of the Paatusoq syenite on the east coast of Greenland (Garde et al., 2002) increases the probability of other alkaline complexes lying beneath the Inland Ice or offshore. Additionally, the dense swarms of benmoreitic and trachytic dykes in the batholith-axial zone (Martin, 1985) may have supplied surface eruptions of 'flood' benmoreite and trachyte lavas on a large scale within an evolving volcano-tectonic graben (Martin, 1985; Macdonald and Upton, 1993).

From the petrogenetic assumptions, generation of the great volumes of salic magma involved in the Gardar Province implies the former presence of much greater volumes of antecedent basaltic magmas. As will be clear from previous sections, these parent magmas would have been produced at intervals of time over the whole 140 Ma or more of the Gardar period. These conclusions lead to the necessity of postulating the involvement of very large volumes of mantle peridotite in the evolution of the Gardar magmas. Consideration of the Ilímaussaq Complex alone, with its extraordinary concentrations of elements such as Cl, Nb, Be (Bailey et al., 1981, 2001; Sørensen, 1997) as ultimate products from fractional melting and crystallisation processes from peridotitic sources, supports the enormity of the requisite source-rock volumes.

12.2. Palaeomagnetism and palaeogeography

Palaeomagnetic results from the three episodes of Gardar magmatism record the earlier limb of an apparent polar wandering loop. This limb is referred to as the Gardar Track (Piper, 1992, 1995) and the return path of the loop is the Keweenawan Track ($\sim$ 1110–1000 Ma), which has been known from studies in North America since the 1960s. The Gardar Track shows that Greenland lay $\sim 10^\circ$ from the palaeoequator during the early part of the Gardar interval before drifting rapidly into higher latitudes, attaining $\sim 72^\circ$ by the time of emplacement of the Ilímaussaq Complex. The most recent interpretation of Neoproterozoic APW paths suggests that Greenland was then near to the South Pole.

12.3. Affinities with North American anorthositic associations

There are clearly similarities between the inferred sub-Gardar anorthosites and the North American massif-type anorthosites (Upton, 1964; Bridgwater,

1967; Bridgwater and Harry, 1968; Halama et al., 2002). The estimated extent of the unseen Gardar anorthositic complex is comparable to the ~19,000 km^2 occupied by the Nain anorthositic complex, Labrador (Hill, 1988). The mid-Proterozoic anorthosite intrusions of Labrador and Quebec have been related to extensional tectonics and failed rifts (e.g. Emslie, 1978; Morse, 1982) as was the case with the Gardar.

The alkaline salic Gardar magmas typically possess strong Sr- and Eu-depleted signatures, attributed to extensive fractionation of plagioclase from the parental magmas. The tendency to peralkalinity, common in the Gardar intrusions, is regarded as an example of the 'plagioclase effect' (Bowen, 1945) and is also ascribed to extended plagioclase fractionation. A genetic link between the hypothesized anorthosite body and the syenitic rocks was proposed by Bridgwater and Harry (1968); subsequent data have reinforced this conclusion, indicating that the more evolved salic melts were probably residual melts that arose from beneath the anorthosites by virtue of their low density and high fluidity (Upton, 1996).

12.4. Mantle plumes and relationship to large-scale Mesoproterozoic events in North America

It is unlikely that the Gardar 'parent' magmas arose solely from adiabatic depressurisation of the underlying mantle during extensional stresses. As indicated above, trace element and isotopic data indicate that the Gardar magmas originated from heterogeneous lithospheric mantle rather than from the asthenosphere. We conclude that the requisite energy was probably derived from convection of the underlying asthenosphere and that three (or more) episodes of mantle plume activity may have been involved. The focal point of these plume 'hot-spots' may have been remote from the Gardar Province while the mechanically weak lithosphere close to the craton margin, undergoing attenuation, provided opportunities for mantle melting and magma ascent (cf. Thompson and Gibson, 1991).

The time interval between the early Gardar events (>1288 Ma) and the mid-Gardar events terminating at ~1275 Ma straddles the great Laurentian rifting event that produced voluminous mafic magmatism in North America, in the interval 1280–1265 Ma. The most widespread expression of this rifting is that of the Mackenzie Igneous Events in NW Canada including the 1272–1265 Ma Mackenzie dyke swarm (LeCheminant and Heaman, 1989; Heaman, unpublished data), 1269 ± 1.1 Ma Muskox layered mafic intrusion (LeCheminant and Heaman, 1989; French et al., 2002) and the Coppermine River flood basalts (LeCheminant and Heaman, 1989). Other correlative mafic dyke swarms in NE Canada include the LP Nain dykes, which yield U–Pb baddeleyite ages of 1280–1277 Ma and which are on-strike with the Gardar BD_0 dykes in pre-drift reconstructions (Buchan et al., 1996) and the NE-trending 1273 ± 1 Ma Harp dyke swarm in Labrador (Cadman et al., 1993). Buchan et al. (1996) also noted a possible correlation between the ENE- and NE-trending dolerite dykes of the Nunarssuit–Isortoq Zone and the 'Harp' dykes in Labrador.

Synchronous mafic magmatism in the Baltic Shield of Finland (Suominen, 1991) indicates that this mafic magmatism was a trans-Laurentian phenomenon. The mid-Gardar magmatism including the 1284–1275 Ma BD_0 dyke swarm and the 1275 Ma Kûngnât gabbro could be early mafic expressions of this large-scale rifting event.

Initiation of the great 'Midcontinent Rift' in North America involved the outpouring of the copious Keweenawan flood basalts, beginning at ~1109 Ma and continuing for 23–1086 Ma (e.g. Davis and Paces, 1990 and references therein). In addition to the large volumes of basaltic magma, numerous ~1.1 Ga alkaline intrusions were emplaced at the margins of the rift (e.g. Heaman and Machado, 1992 and references therein). However, these intrusions, which include a variety of rock types (e.g. alkaline gabbro, syenite, carbonatite), are more than 30 Ma younger than the youngest alkaline intrusion in the Gardar Province. Although there are some similarities in the geology of the Keweenawan and Gardar rifts, the precise U–Pb geochronology now available demonstrates that they resulted from distinct rifting events in Laurentia separated in time by 30–40 my. The closest temporal equivalents in North America to the latest Gardar magmatism may be the ultramafic lamprophyre dykes intruded in the Lake Superior region at about 1145 Ma (Queen et al., 1996) and the 1141 ± 2 Ma Great Abitibi diabase dyke (Krogh et al., 1987).

13. Summary

Continental rifting in the Proterozoic attained three major culminations at 2.0–1.8, 1.2–0.9 and 0.8–0.6 Ga. The Gardar rifting, which clearly occurred in the second of these, should be referred to the 'passive' rather than 'active' categories of rift systems (Green, 1992 and references therein).

While nowhere in southern Greenland is the anorthositic complex exposed, it can scarcely be doubted that such a complex, developed at lower to middle crustal depths, played a seminal role in the formation of the Gardar alkaline magmas. The anorthosite may be regarded as an integral part of the zone of mid-Proterozoic anorthosite intrusions (with ages from ~ 1800 to ~ 970 Ma), traceable southwest across North America, some 4800 km, from Labrador to California (Scoates and Chamberlin, 1995). It differs from most of the North American instances in being (a) younger, (b) more alkalic and (c) shallowly eroded, preserving remarkable sequences of syn-rift volcanic and sedimentary rocks.

Since the tectonics operating throughout Gardar time were extensional in nature, the hypothesis that the anorthosites formed as a result of crustal underthusting (Duchesne et al., 1999) appears inapplicable and, on the basis of the data presently available, a mantle rather than a lower crustal origin remains the preferred explanation. The intimate relationship between faulting and magmatism in space and time throughout the period implies that source melting was triggered by pressure relief attending the faulting.

The geochemistry of the Gardar basaltic rocks suggests that involvement of the lithospheric mantle was critical and it is proposed that the basaltic magmas were generated within variably metasomatised lithospheric mantle. The observation that, over some 200 Ma, a suite of igneous rocks with such close petrographic and geochemical affinities with plutons whose internal structures have remarkable similarity was produced points to their having originated from a relatively uniform source. Over so long a time span this source could only be non-convecting lithospheric mantle. Whereas the remarkable association of carbonated ultramafic lamprophyres and carbonatites that made episodic appearance throughout much of the Gardar period may have been due to large-fraction melting of fusible, highly enriched, lithospheric metasomatites, the alternative hypothesis that they denote very small fraction 'primary' melts of asthenospheric origin, cannot yet be discounted.

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